

Machine Learning

Supervised Learning

Supervised learning uses labeled training data to learn a relationship between input features and target outputs. Classification predicts categories, while regression predicts continuous numerical values.

Unsupervised Learning

Unsupervised learning works with data without labeled target values. Clustering groups similar observations, while dimensionality reduction represents data using fewer features.

Model Evaluation

Common classification metrics include accuracy, precision, recall, and F1 score. Regression can be evaluated using mean squared error, mean absolute error, and R-squared.